Date 24/8/26
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24/8/26 :-

```
main()
{   int A,B,C;
    A = 15;
    B = 7;
    C = A + B;
    printf("Add");
    printf("%d", C);
}
```

① int A → Declaration → A is a variable containing value of int type → %d

② A = 15 → Definition (15 is value of A)

③ int i = 7; → initialization

25/8/26 :-

```
main()
{   float n1, n2;
    float Sum;

    n1 = 7.1;
    n2 = 9.9;

    printf("Add");
    printf("%f", Sum); }
```

n1 = 7.1 | n2 = 9.9 | Sum

```
printf("Sum of %f", n1, n2, Sum)
printf("Sum of %f and %f is %f", n1, n2, Sum)
```

(n1) (n2) (Sum)

Sum of 7.3 and 6.2 is 13.5

Size of int = 2 Byte.
" " float = 4 Byte.
char = 1 Byte

31/8/26:-

```
main ()
{ float div;
{ int A, B, Sum, sub, mul; div;
    A = 1;
    B = 2;
    Sum = A + B;
    sub = B - A;
    mul = A * B;
    div = B/A;
printf("Sum of %d and %d is %d", A, B, Sum);
printf("sub of %d and %d is %d", B, A, sub);
printf("mul. of %d and %d is %d", A, B, mul);
printf("div. of %d and %d is %f", B, A, div);

}
```

Table of 4:-

```
main ()
{ int A, n1, n2, n3, n4, n5, n6, n7, n8, n9, n10; x1, x2,
                x3, x4, x5, x6, x7, x8, x9, x10;
  A = 4;
  n1 = 1;
  n2 = 2;
  n3 = 3;
  n4 = 4;
  n5 = 5;
  n6 = 6;
  n7 = 7;
  n8 = 8;
  n9 = 9;
  n10 = 10;
```

For Optimization of Code:- ① Storage
② Line of Code (LOC)
③ Execution (Output).

```
x1 = A * n1;
x2 = A * n2;
x3 = A * n3;
x4 = A * n4;
x5 = A * n5;
x6 = A * n6;
x7 = A * n7;
x8 = A * n8;
x9 = A * n9;
x10 = A * n10;

~~printf (x1, x2, x3, x4, x5, x6, x7, x8, x9, x10);~~
printf ("%d, %d, %d, %d, %d, %d, %d, %d, %d, %d", x1, x2, x3, x4,
        x5, x6, x7, x8, x9, x10);
}
```

⇓

```
main()
{ int A, n;
   A = 4;
   n = A * 1;
  printf ("%d", n);
   n = A * 2;
  printf ("%d", n);
   n = A * 3;
  printf ("%d", n);
   n = A * 4;
  printf ("%d", n);
   n = A * 5;
  printf ("%d", n);
```

```
n = A * 6;
printf("%d", n);
n = A * 7;
printf("%d", n);
n = A * 8;
printf("%d", n);
n = A * 9;
printf("%d", n);
n = A * 10;
printf("%d", n); }
```

1/9/26)

```
scanf("%d", &n1)
```

address of.

(

```
main()
{   int n1, n2, Sum;

    scanf("First Digit %d", &n1)
    scanf("Second Digit %d", &n2)

    Printf("n1");
    scanf("%d", &n1);
    Printf("n2");
    scanf("%d", &n2);
    Sum = n1 + n2;
    Printf("Sum is %d", Sum);
}
```

Date
Page

==, <, >, <=, >=

2/9/26)

```
i = 2;

if (i == 1)
{
    printf("one");
}
else
{
    printf("not one");
}
```

T.B

F.B

```
if (<Condition>)
{
    True Block
}
else {False Block}
```

<Condition> replaceable.

7/9/26)

lib. files → #include <stdio.h> ← Pre Processor; stdio → Std. input output

```
void main()        ← func"
{   int n1, n2;    ← Variable declaration.

    printf("Enter n1 and n2");
    scanf("%d %d", &n1, &n2);
    if (n1 > n2)
    {
        printf("n1 is greater");
    }
    else
    {
        printf("n2 is greater");
    }
}
```

func" ← printf

scanf: %d %d ← formula specifier; &n1, &n2 → add of

For	n_1	n_2
	7	3 ✓ Works
	3	7 ✓ Works.
	5	5 ✗ Don't work.

I) if ($n_1 == n_2$)
{ printf("n1 and n2 are equal")
}
if ($n_1 > n_2$)
{ printf("n_1 is greater than n_2")
}
if ($n_2 > n_1$)
{ printf("n_2 is greater than n_1")
}

For n_1 n_2
7 3 ✓
3 7 ✓] work.
5 5 ✓

II) if (n1 == n2)
{
① eq.
}
else
{
if ($n_1 > n_2$)
{ n_1
}
else { n_2
}
}

&& → AND ; || → OR
!= ⇒ Not equal.

~~Divisible by no. x~~ ~~if (x == 0)~~
~~{ printf("Divisible")~~

A no. divisible by other no:-

```
{ a, x, i;
  printf("Enter the no.");
  scanf("%d", &a);
  printf("Enter the divisor");
  scanf("%d", &i);
  x = a%i;

  if (x == 0)
     { printf("Divisible");
     }
  else { printf("Not Divisible");
       }
  return 0;
}
```

8/9/26 :- Greatest of 3 no :-

n_1	n_2	n_3
2	5	8
8	2	5
5	8	2

```
# include <stdio.h>
int main()
{ int n1, n2, n3;
  printf("Give 1st No. :- ");
  scanf("%d", &n1);
  printf("Give 2nd No. :- ");
  scanf("%d", &n2);
  printf("Give 3rd No. :- ");
  scanf("%d", &n3);

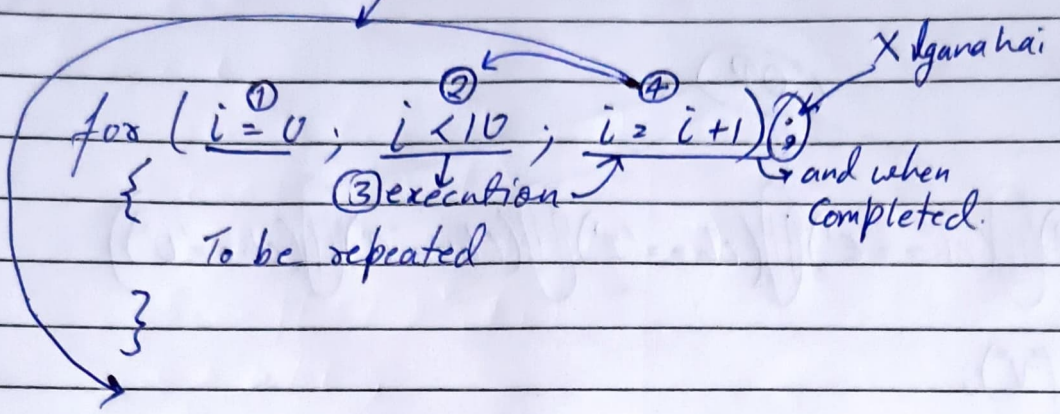
if (n1 > n2)
 { if (n1 > n3)
     { printf("1st is gotr");
     }
 }
if (n2 > n1)
 { if (n2 > n3)
     { printf("2nd is gotr");
     }
 }
if (n3 > n1)
 { if (n3 > n2)
     { printf("3rd is gotr");
     }
 }
return 0;
}
```

```
if (n1 > n2 && n1 > n3)
 { printf(1st)
 }

if (n2 > n1 && n2 > n3)
 { print("2nd")
 }

if (n3 > n1 && n3 > n2)
 { printf("3rd")
```

14/9/26 - Loops - For While Do While.



⇒ Escape Sequence → Like \n.

15/9/26 -

```
for(i=1 ; i<16 ; i=i+1)
  {
    printf("%d", i);
  }
```

Switch - Case -

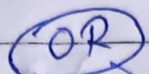
```
Switch (x)
{
  case 0...50 :
    ——— break;
  case 1
    ——— break;
}
```

x = x, Range.

Agar nhi lgaye, To Case True uske aage wale sare True.

16/9/2021-

```
char ch;
ch='A';
if((ch=='A')||(ch=='E')||(ch=='I')||(ch=='O')||(ch=='U'))
  {
    pf("V");
  }
```



s | e
2 | 15

How the Code runs:-

```
for(i=s; i<=e; i++)
  {
    if(i%2==0)
      pf()
      cnt=cnt+1   <- for Counting.
  }
```

i++ ⇒ i = i+1

i-- ⇒ i = i-1

4321

4 × 1000
3 × 100
2 × 10
1 × 1

1234

1 × 1000
2 × 100
3 × 10
4 × 1

$x = f(2,3)$ $x = y$; $x = x+2$; Exp

28/9/26 :-

$x = 2$; True/N.Zero. False/Zero.

```
if (x = 1)
{
}
```
instead of "0" every no is N.Zero.

```
if (x = 0)
{
}
```

```
if (!(x>50))
```
F ↑ I

```
if (!(x<50))
```
I F

```
x = 0;
if (x)
```

29/9/26 :-

```
void addf();
```
Function Dectaration.

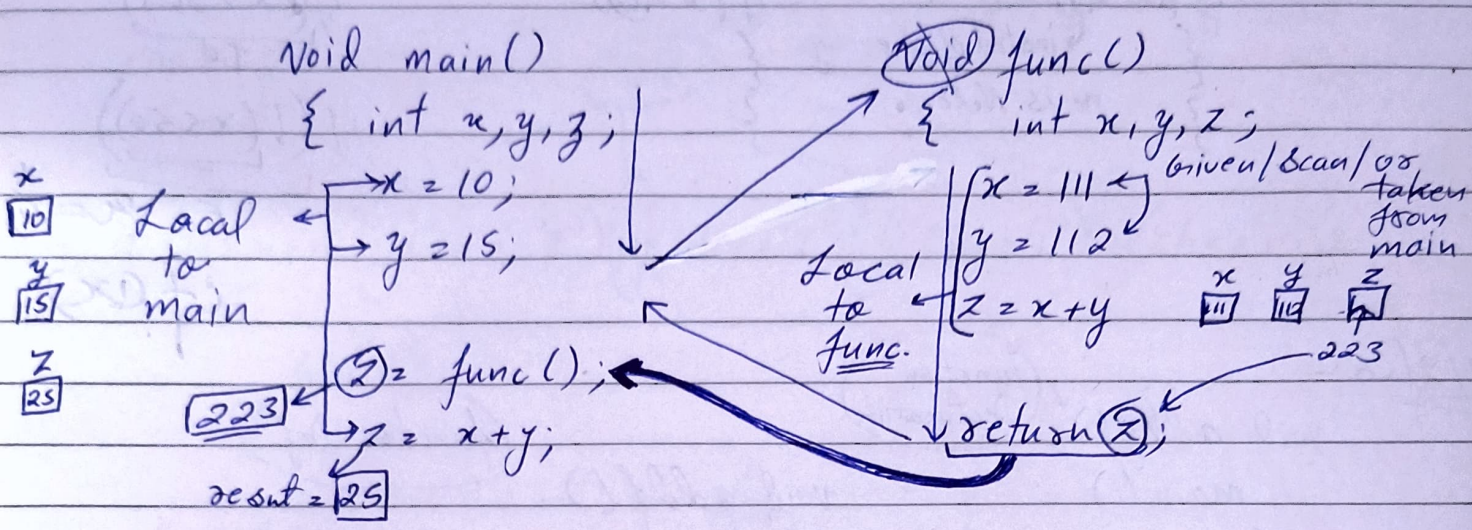
```
main()
{
   addf();
}
```
Function call

```
void addf()
{
   int n1, n2, R;
   printf("N1");
   scanf("%d", &n1);
   R = n1 + n2;
   printf("%d", R);
}
```
Function Def"

Syntax of Def" :-

```
<dt> <fname> ()
{
   fn body.
}
```

30/09/26:-



06/10/26:- int fpoint();

```
main()
{ int i, j, x;
  i = 7;
  j = 21;
  x = fpoint(i, j);
}
```

Can be no. of variables.

```
int fpoint(int x, int y)
{ int z;
  z = x + y;
  return z;
}
```

7/10/26:- While loop:-

Zero Value → False; Non-zero Value → True or more

```
i = 0;
while (i < 10)
{ pf(i);
  i++;
}
```

```
while ('A')
{ S1
  S2
}
```

Infinite Loop

```
i=0;
for(___; i<=10; i++)
{ ___ ___ ___
}
```
Will Run

```
i=0;
for(___; ___; i++)
{ ___ ___ ___ ___
    if(i<=10);
    { break;
    }
}
```
Will Run

```
i=0;
for(___; ___; ___)
{ ___ ___ ___ ___ ___
    if(i<=10)
    { break;
    }
  i++;
}
```
Will Run

→ if we remove initialization, condition, change in "for" and put them either before or after the loop It will still run normally.

Table → Array → Matrices.

Secondary.

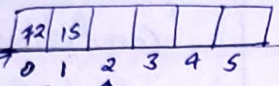
14/10/2026 – Array.

Primary.

int, float, char.

int i = 7

int ai [6] ⇒



Math ↓ Matrices

Declar"

Def"

ai [0] = 72

ai [1] = 15

scanf("%d", &ai[2]);

initialization.

Agar int ai [6] = {1, 2, 3, 4, 5}

